

The Full Strong Completeness of $L^p(T)$ Is Settled

Literature-answer packet for arXiv:2304.07251
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Status. Already answered in the literature. Azouzi's later arXiv:2512.11809 proves strong completeness for nets in $L^p(T)$ for every $p \in [1, \infty]$, under the same conditional-Riesz-triple hypotheses.

1 The source question

Azouzi's 2023 paper proves *sequential* strong completeness of $L^p(T)$ for every $p \in [1, \infty]$. It explicitly distinguishes this from full strong completeness, where arbitrary Cauchy nets rather than only sequences must converge. The paper records that the full problem was known only at $p = 2$ and $p = \infty$ [1, p. 2].

established the strong sequential completeness of the space $L^1(T)$. Their proof is not an adaptation of the proof for classical space $L^1(\mu)$ because it is no longer true that a strong Cauchy sequence (x_n) has a subsequence (y_n) such that the series $\sum y_n$ is order convergent. Our result completes their work and establishes that all spaces $L^p(T)$ are sequentially strong complete for $p \in [1, \infty]$. However, the full completeness of these spaces is still an open problem, and has only been proven for two values of p . The first case, $p = \infty$, as a relatively straightforward proof and can be found in [16], while the second case (for $p = 2$) is much more complex and has recently been demonstrated by Kalauch, Kuo, and Watson in [12]. Their proof is based on a Hahn-Jordan type decomposition which is proved in a separate paper [11].

Figure 1: The open full-completeness problem in arXiv:2304.07251, PDF page 2.

The setting is a conditional Riesz triple (E, e, T) : E is a Dedekind complete Riesz space with weak order unit e , and T is a conditional expectation operator. The vector-valued p -norm is

$$N_p(x) = T(|x|^p)^{1/p},$$

and strong completeness means convergence of every net that is Cauchy for this norm in the order sense.

2 The later answer

The same author returned to this exact problem in the 2025 preprint *Strong completeness of L_p -type vector lattices* [2]. Its Theorem 19 proves, for each fixed $p \in (1, \infty)$, the equivalence

$$L^1(T) \text{ is strongly complete} \iff L^p(T) \text{ is strongly complete.}$$

Combining this with the already established $p = 2$ theorem yields the full finite- p range; the known $p = \infty$ result supplies the endpoint. The paper states the resulting conclusion as Theorem 20.

proof. ■

As a consequence of Theorem 19, and [25, Theorem 4.1] we deduce the following.

Theorem 20 *Let (E, e, T) be a conditional Riesz triple. Then $L^p(T)$ is strongly complete for every $p \in [1, \infty]$.*

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Figure 2: Theorem 20 of arXiv:2512.11809, PDF page 16.

Theorem 1 (Azouzi, 2025). *Let (E, e, T) be a conditional Riesz triple. Then $L^p(T)$ is strongly complete for every $p \in [1, \infty]$.*

This is precisely the property left open in arXiv:2304.07251. It concerns arbitrary nets, not merely sequences, and it includes all formerly missing values of p .

3 Match audit

The source and answer align without a change of scope:

- both assume a conditional Riesz triple (E, e, T) ;
- both use the same spaces $L^p(T)$ and their natural vector-valued norms;
- the source asks for full strong completeness of nets;
- the later theorem asserts full strong completeness for every $p \in [1, \infty]$.

Accordingly, the open question should be marked completely answered in the literature, rather than treated as a target for a new proof.

4 Human review recommendation

This is a direct theorem-to-question match and needs only light review. Confirm the definition of “strongly complete” in arXiv:2512.11809 (every strong Cauchy net converges) and the common conditional-Riesz-triple hypotheses. Theorem 20 then closes the source question verbatim.

References

- [1] Youssef Azouzi, *Uniform Integrability in vector lattices and applications*, arXiv:2304.07251, 2023. <https://arxiv.org/abs/2304.07251>.
- [2] Youssef Azouzi, *Strong completeness of L_p -type vector lattices*, arXiv:2512.11809, 2025. <https://arxiv.org/abs/2512.11809>.